



Quantifying Mutual Intelligibility using Language Models: Evidence from Turkic Languages

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Research Motivations

Mutual intelligibility (MI) refers to how well speakers of language A can understand language B through *lexical* and *grammatical* correspondences.

Current **computational approaches** estimate MI from lexical or phonetic distances based on *hand-curated cognate lists*.

GAP: Can a language model (LM) capture the MI gradients of closely or distantly related languages?

Hypothesis

LM trained on a *source language* and fine-tuned on *more closely related target languages* will exhibit **greater generalizability** than when fine-tuned on *distantly related ones*.

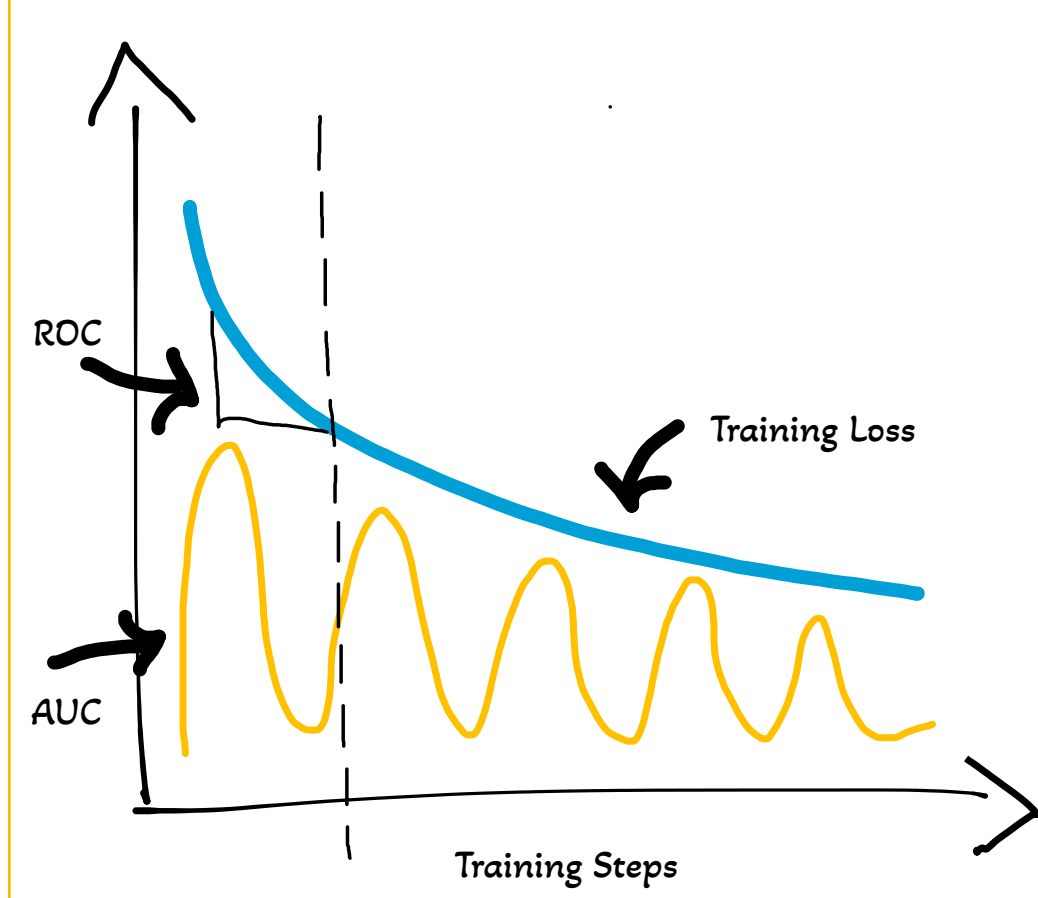
How to measure model's performance?

Model metrics

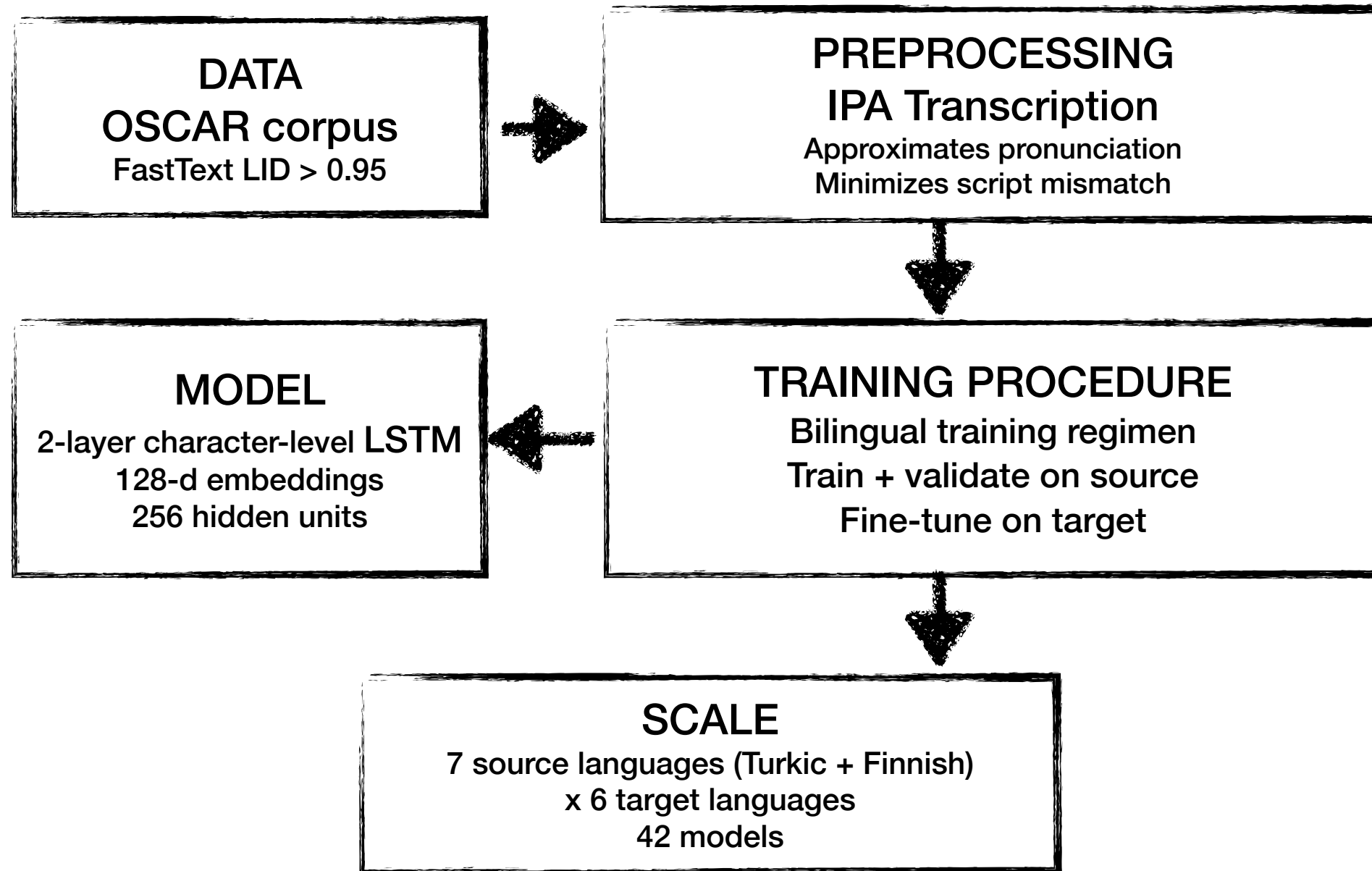
- CE training loss:** prediction error during training
- AUC learning dynamics:** overall learning across training
- ROC:** rate of learning

Expected pattern

- Closer language pairs → **lower CE loss, smaller AUC**
- More distant language pairs → **higher rate of change**



Approach

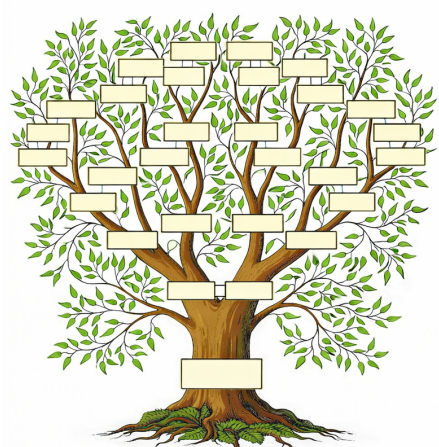


MI among Turkic languages

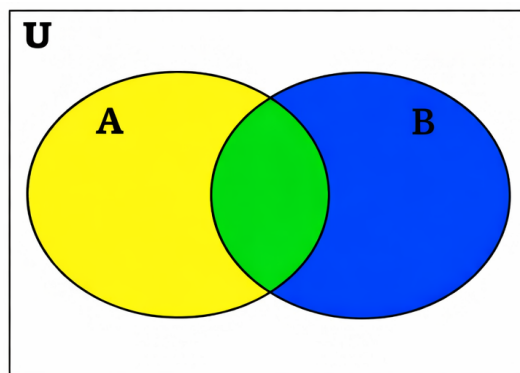
Language	Branch	Sentence in original script	IPA transcription
Turkish	Oghuz	(Ben) (bir) kitap okuyorum.	(ben) (bir) kitap okujorum.
Azerbaijani		(Mən) kitab oxuyuram.	(mən) kitab oxujuram.
Kazakh	Kipchak	(Мен) кітап оқып жатырмын.	(men) kitap oqap zatərmən.
Kyrgyz		(Мен) китеп окуп жатам.	(men) kitep okup zatam.
Uzbek	Karluk	(Men) kitob o'qiyapman.	(men) kitob oqijapman.
Uyghur		(Men) kitab oquwatimen*	(mən) kitab oquwatimən.

Table 1: MI among Turkic languages illustrated by the sentence: *I am reading a book.*

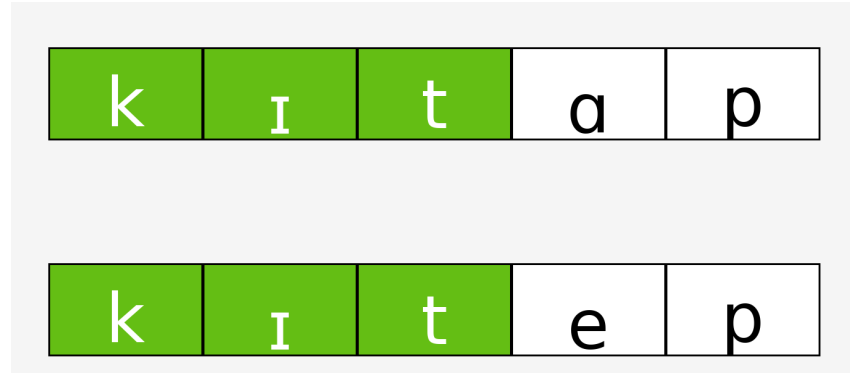
What is driving MI?



Tree distance?

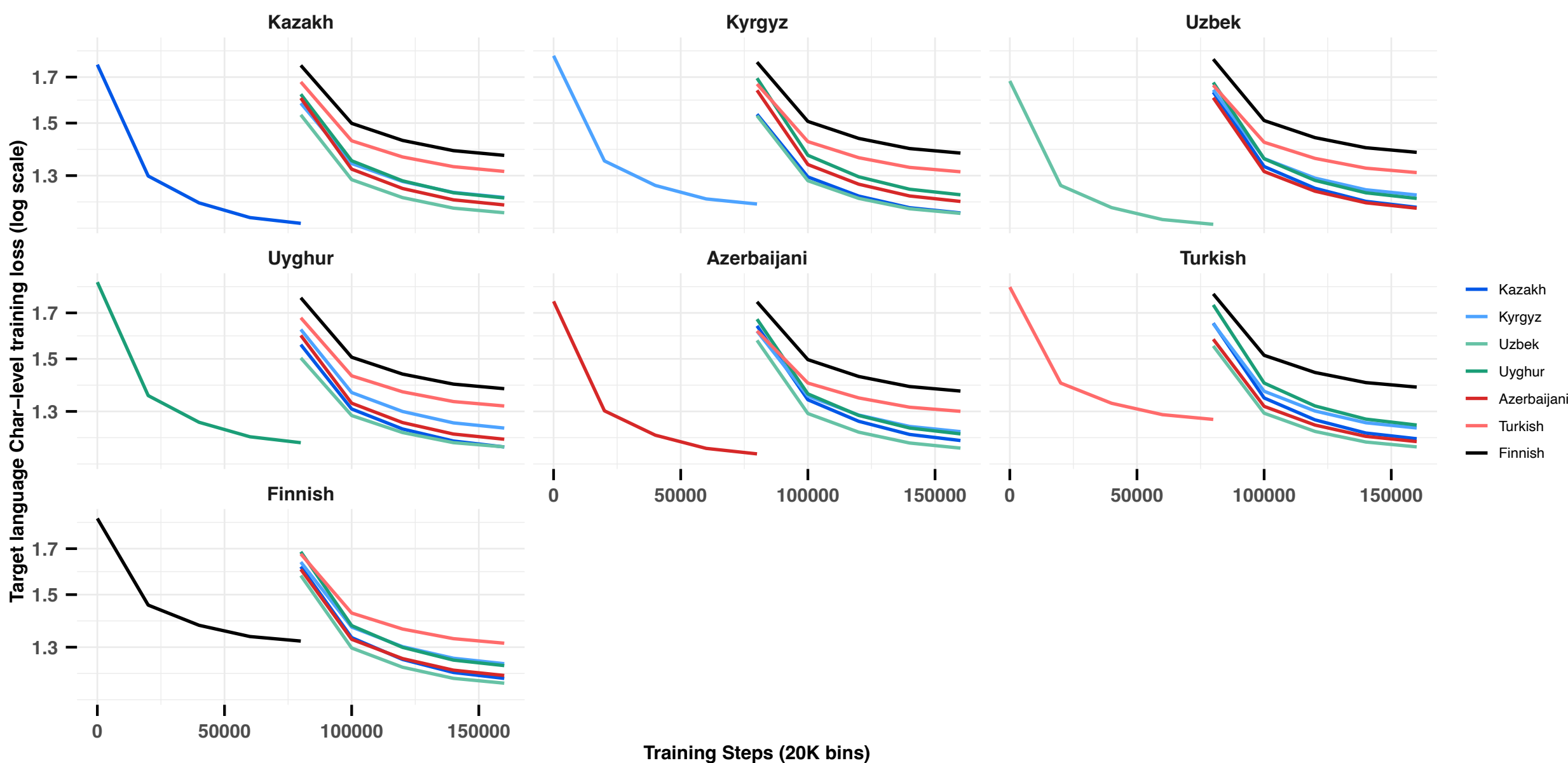


Lexical overlap?



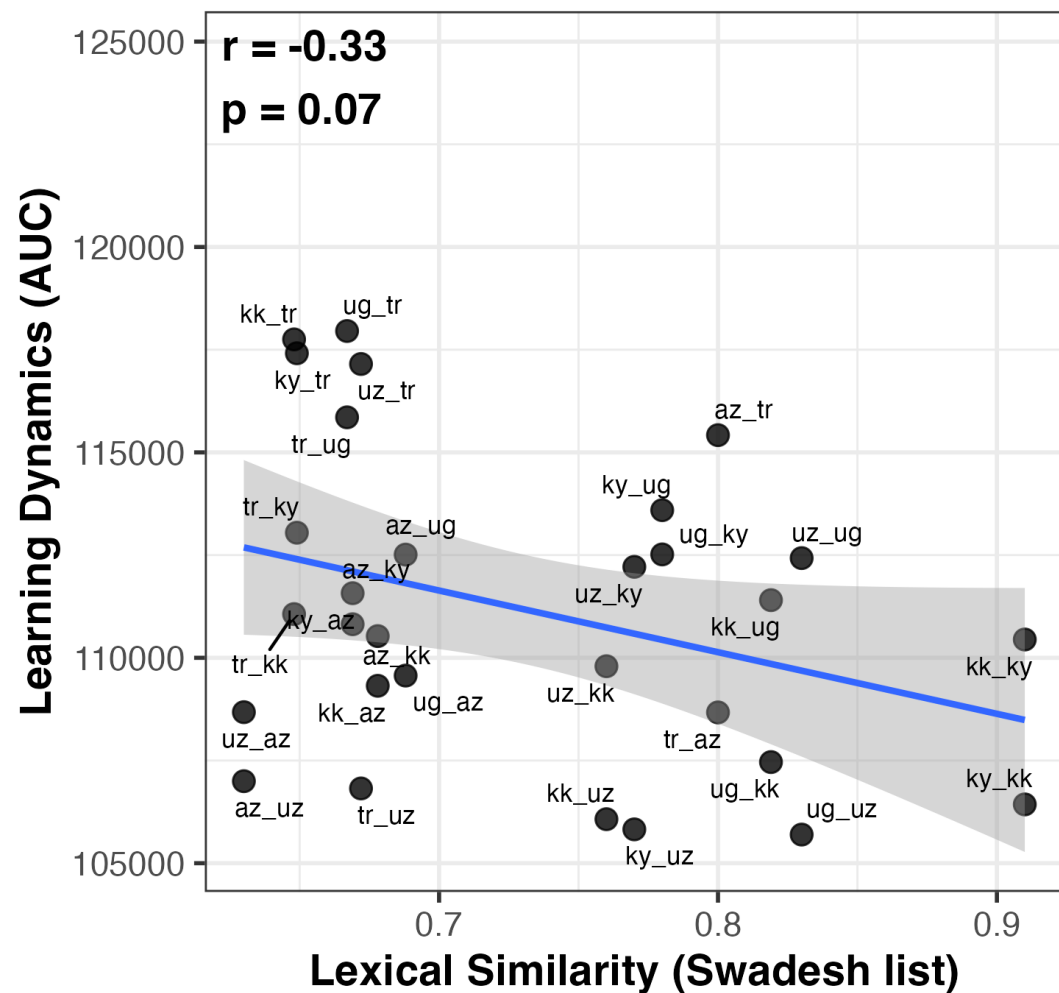
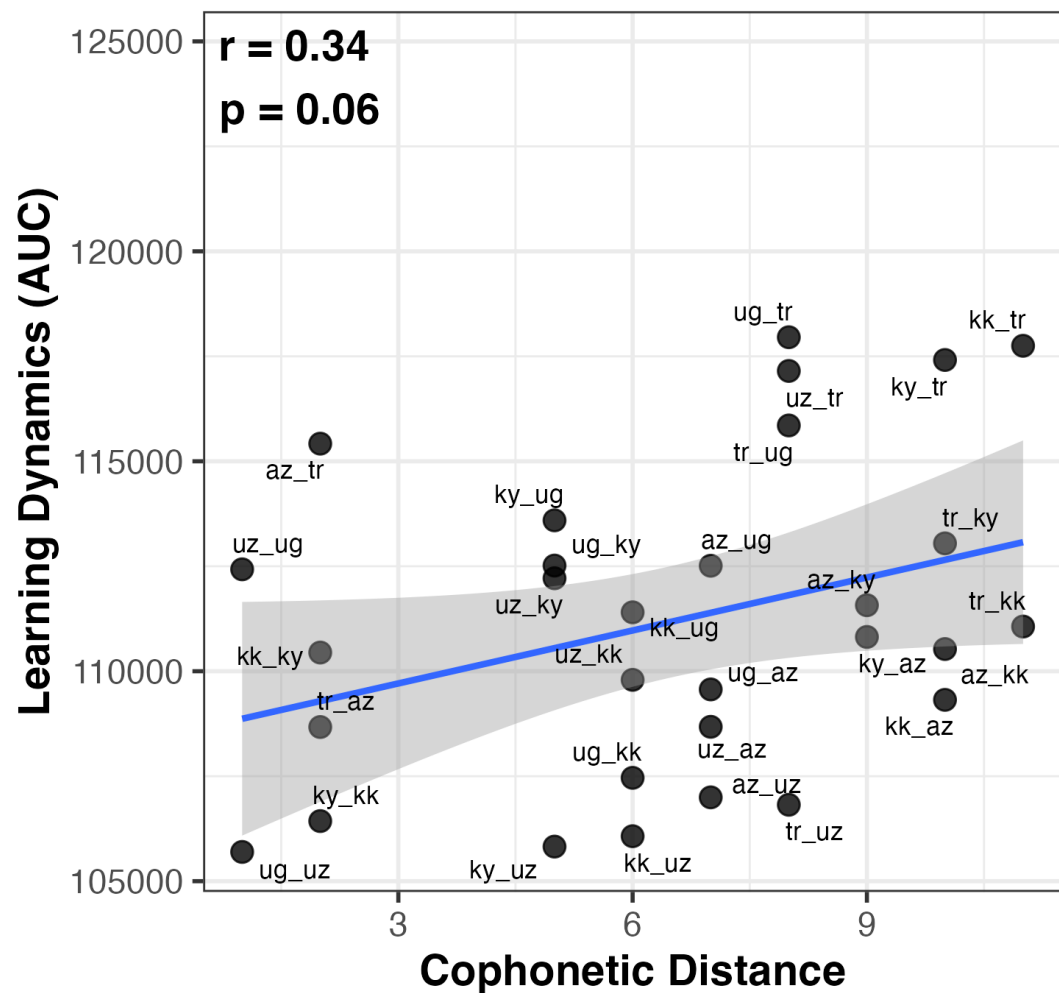
Trigram overlap?

CE Loss by Source vs Target Language



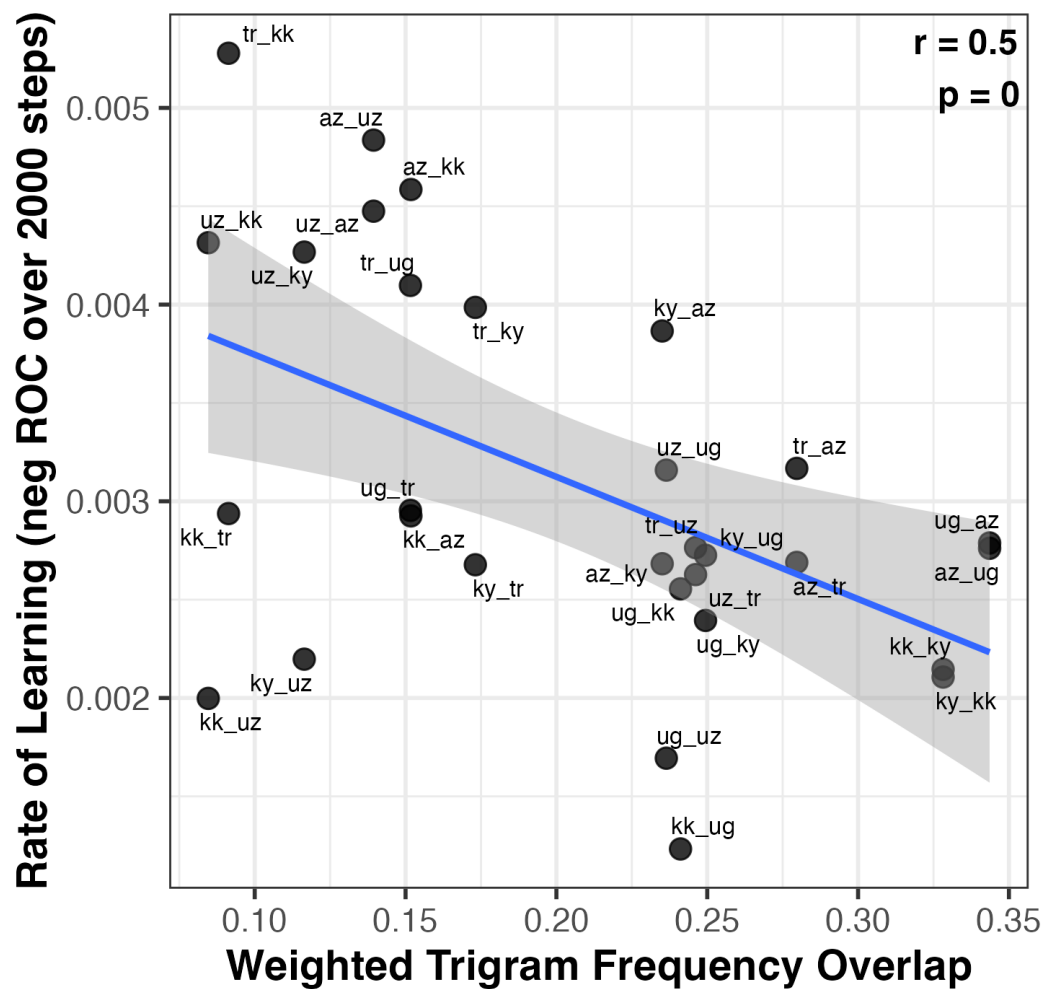
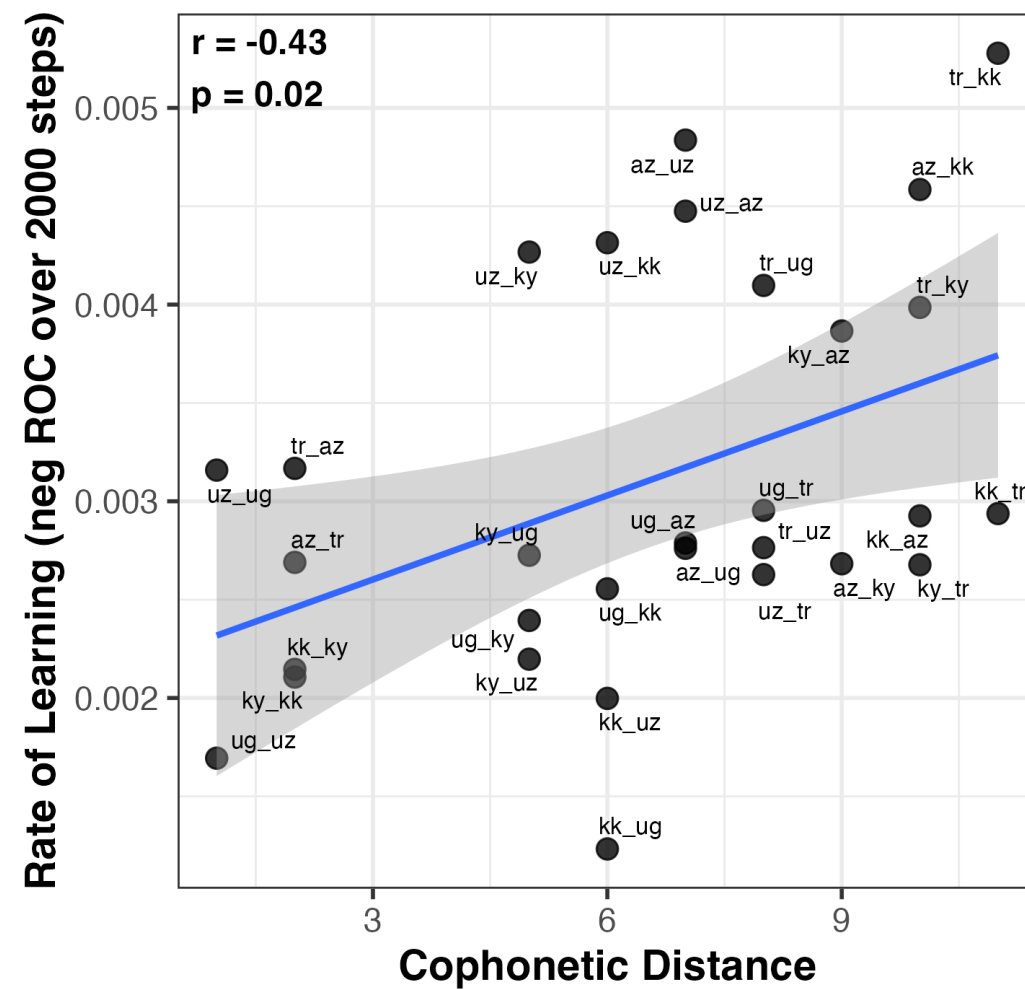
CE patterns broadly tracked known MI gradients: closer Turkic pairs showed smaller loss spikes, faster convergence, and lower training loss. Transfer was better from Kyrgyz to Kazakh than to Turkish, from Turkish to Azerbaijani than to Kazakh, and from Uyghur to Uzbek than to Kazakh. **Finnish showed higher CE loss across all pairs.**

Learning Dynamics



As predicted, the AUC pattern qualitatively was lower for closer language pairs and for pairs with greater lexical overlap.

Rate of Learning



Overall, greater language distance predicts greater learning change: more cophonetically distant pairs show higher ROC, whereas pairs with greater trigram overlap show lower ROC.

Summary

[CE ↓ | AUC ↓] Overall transfer: Closer related pairs showed lower CE loss and smaller AUC, consistent with greater generalizability.

[ROC ↑ for distant pairs] More distant pairs showed a higher rate of change in the early stages of training.

[Genealogy > Typology] Finnish results suggest that genealogical relatedness matters more than typological overlap alone.

[Asymmetry > Similarity] Models appear to be sensitive to asymmetric relationships between language pairs.

[Model vs Human] Open question

But do native speakers perceive these similarities the way the model does?

Selected References: [1] Gooskens & Heuven (2021). Mutual Intelligibility. [2] Tamborisev (2001). The phonological distances between Turkic languages based on some phonological features of consonants. [3] Lindsay (2010). Mutual intelligibility among the Turkic languages. [4] Savelyev, A., & Robbeets, M. (2020). Bayesian phylogenetics infers the internal structure and the time-depth of the Turkic language family. [5] Hochreiter et al. (1997). Long short-term memory. [6] Arnett et al. (2025). On the acquisition of shared grammatical representations in bilingual language models.